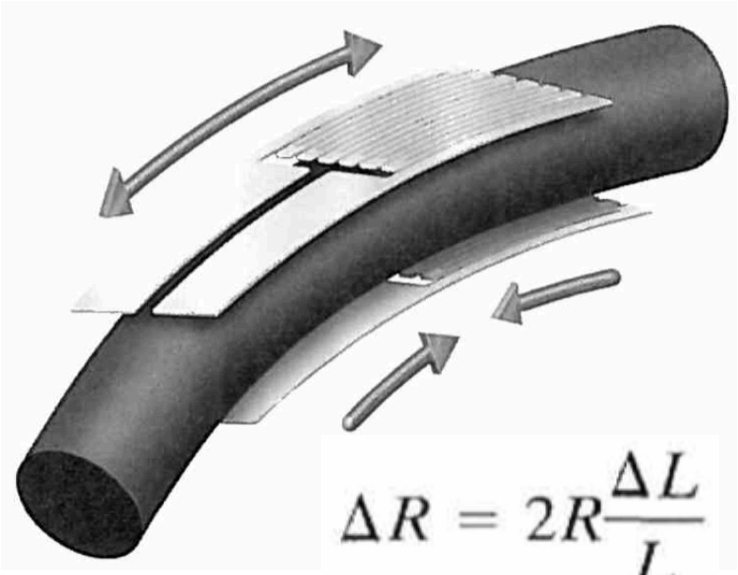
The Operational Amplifier

Qi Xuan
Zhejiang University of Technology
October 2015

Structure

- Operational Amplifier Terminals
- Terminal Voltages and Currents
- The Inverting-Amplifier Circuit
- The Summing-Amplifier Circuit
- The Noninverting-Amplifier Circuit
- The Difference-Amplifier Circuit
- A More Realistic Model for the Operational Amplifier

Strain Gages



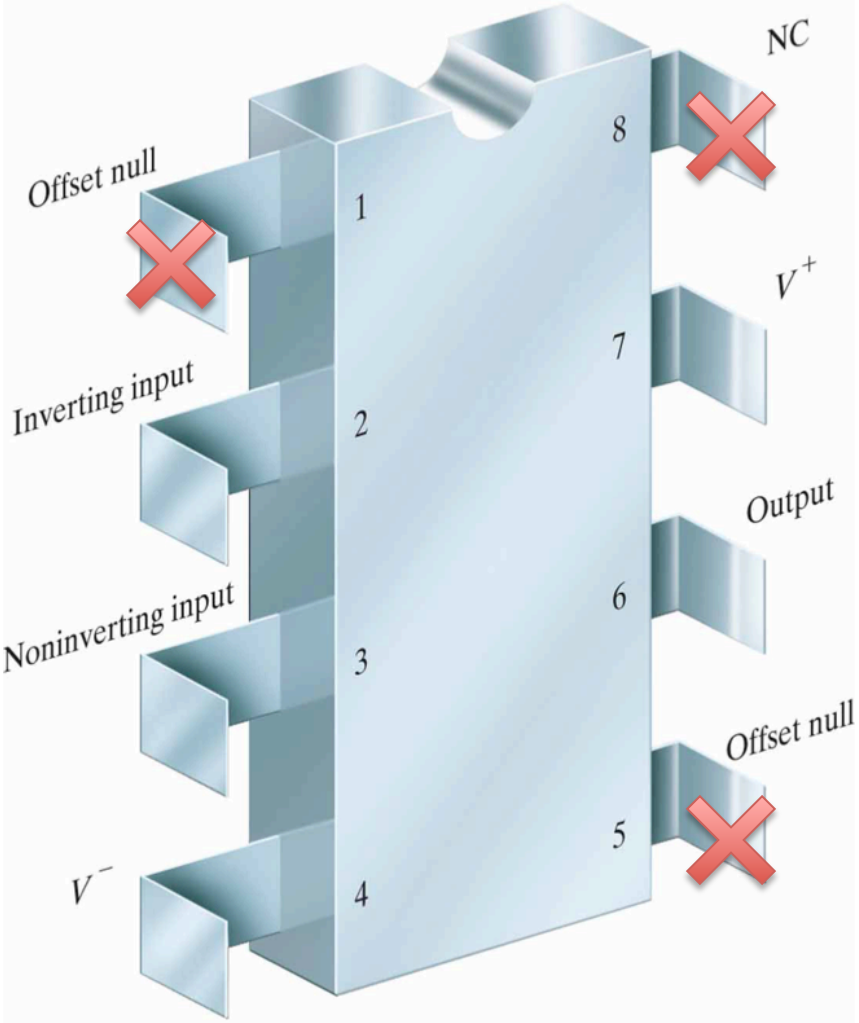
A strain gage is a grid of thin wires whose resistance changes when the wires are lengthened or shortened.

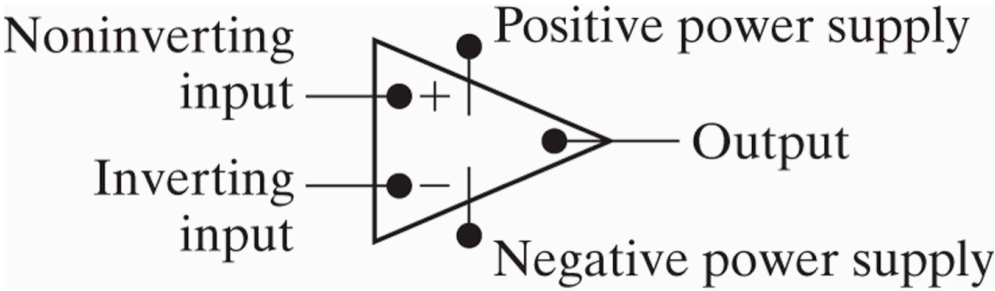
R: the resistance of the gage at rest;
ΔL/L: is the fractional lengthening of the gage (which is the definition of "strain");
2: is typical of the manufacturer's gage factor;
ΔR: the change in resistance due to the bending of the bar.

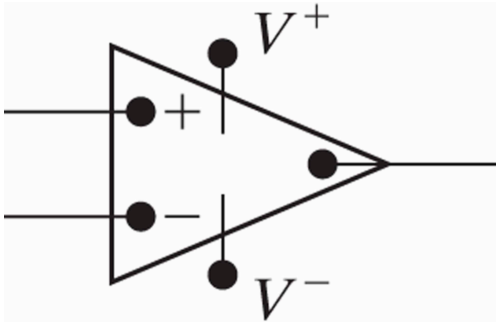
The change in resistance experienced by the strain gage is typically ***much smaller*** than could be accurately measured by an ohmmeter.

In order to make an accurate measurement of the voltage difference, we use an operational amplifier circuit to ***amplify, or increase, the voltage difference***.

Operational Amplifier Terminals

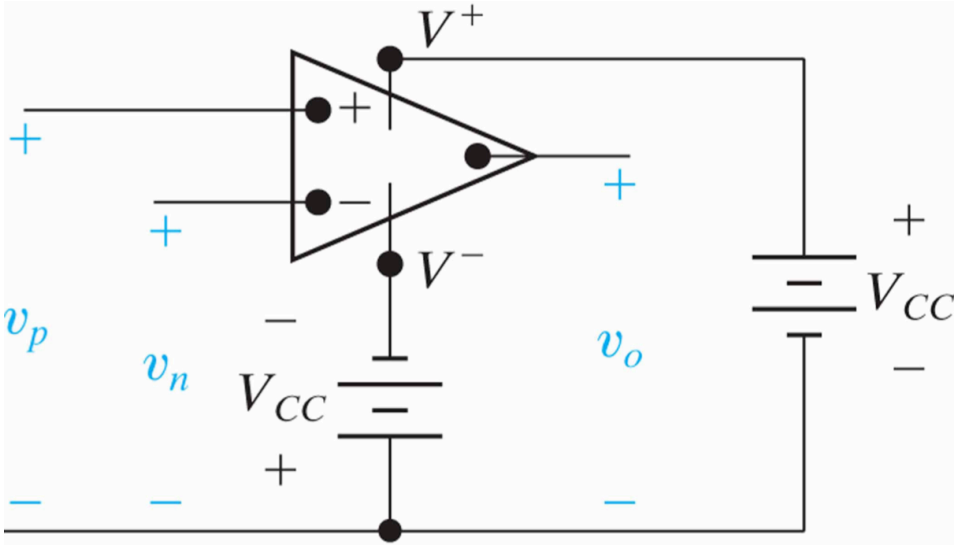


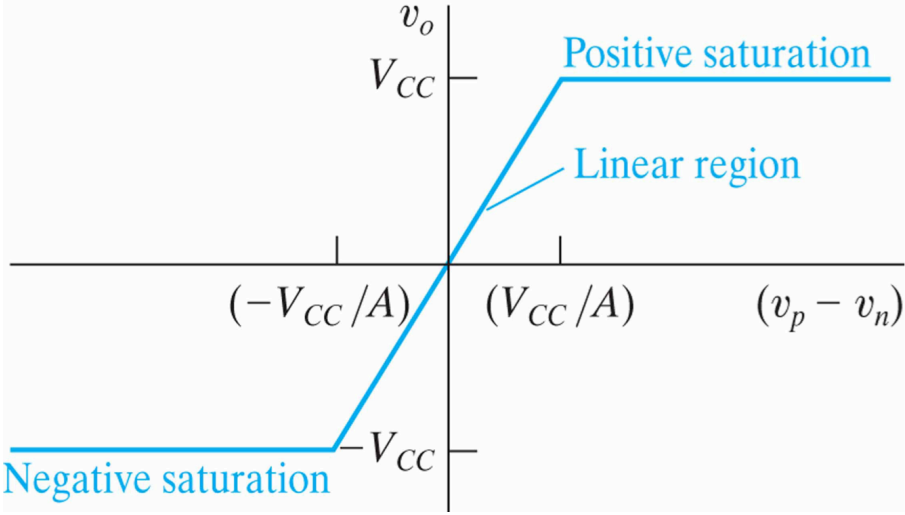




A simplified circuit symbol for an **op amp**

Terminal Voltages





$$v_o = \begin{cases} -V_{CC} & A(v_p - v_n) < -V_{CC}, \\ A(v_p - v_n) & -V_{CC} \le A(v_p - v_n) \le +V_{CC}, \\ +V_{CC} & A(v_p - v_n) > +V_{CC}. \end{cases}$$

Saturate

Linear region

Saturate

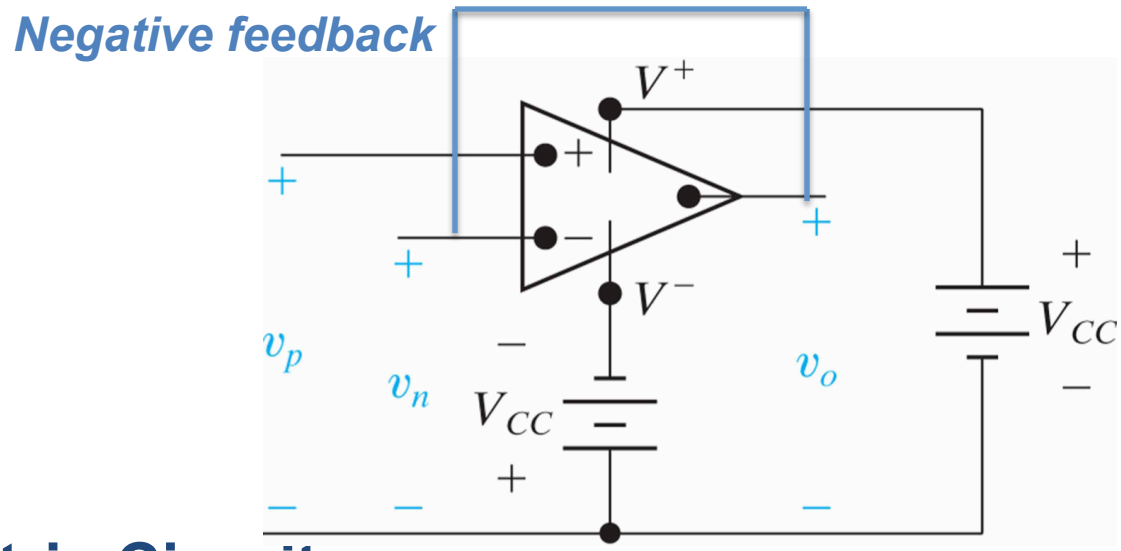
Ideal Op Amp

$A = +\infty$ Linear region

$V_{CC} \leq 20$ V

$v_p = v_n$ Virtual short condition

Maintain the virtual short condition to ensure linear operation:

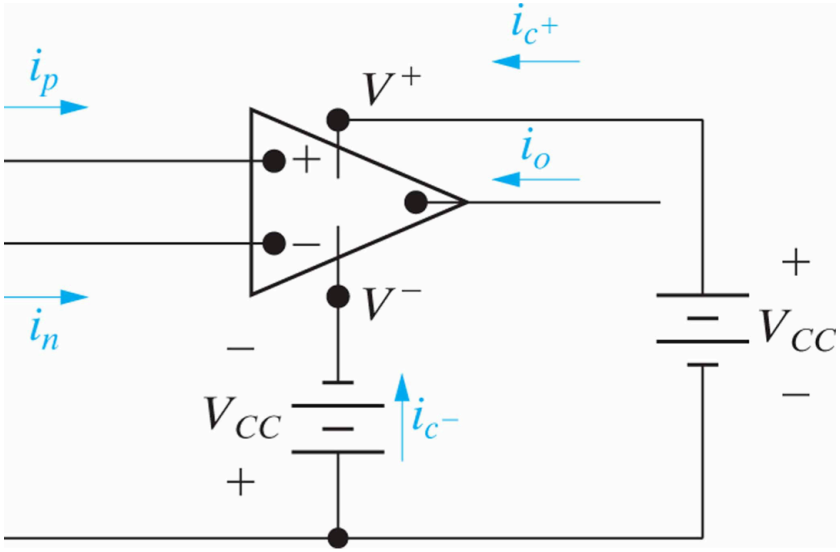


Puzzle

- Even if the circuit provides a negative feedback path for the op amp, linear operation is not ensured. So *how do we know whether the op amp is operating in its linear region?*
- The answer is: we don't know!

- We first assuming linear operation, performing the circuit analysis, and then checking our results for contradictions, if we have $-V_{CC} \leq v_o \leq V_{CC,}$ it's in the linear region, if not, it's saturate!

Terminal Currents



Ideally, the equivalent input resistance is infinite, resulting in the current constraint:

$$i_p = i_n = 0$$

Note that the current constraint is not based on assuming the op amp is confined to its linear operating region as was the voltage constraint.

Kirchhoff's current law: $i_p + i_n + i_o + i_{c^+} + i_{c^-} = 0$

$$i_o = -(i_{c^+} + i_{c^-})$$

Notes

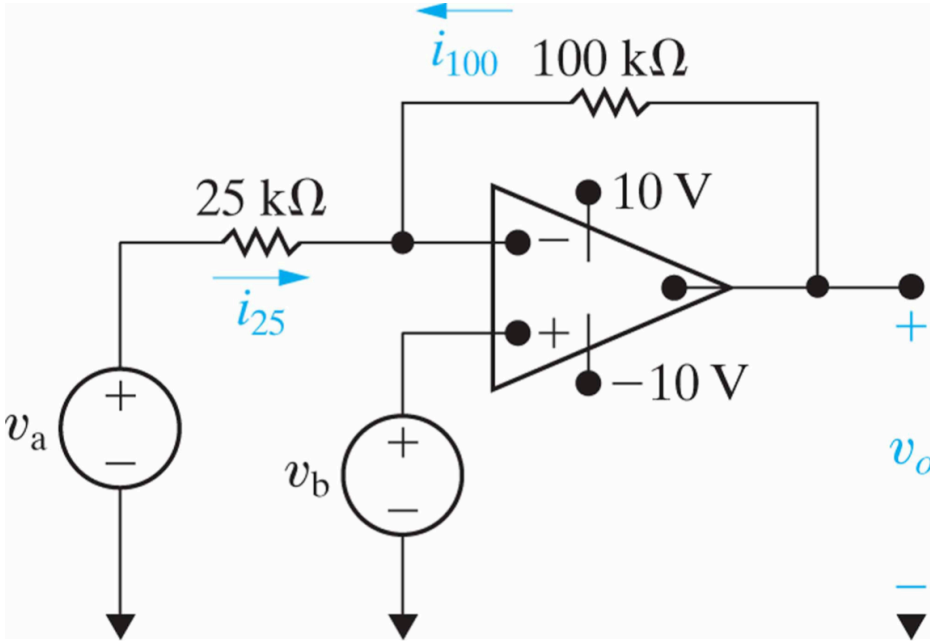
- The positive and negative power supply voltages do *not have to be equal* in magnitude. In the linear operating region, v_o must lie between the two supply voltages.
- Be aware also that the value of A is *not constant* under all operating conditions. For now, however, we assume that it is.

Example #1

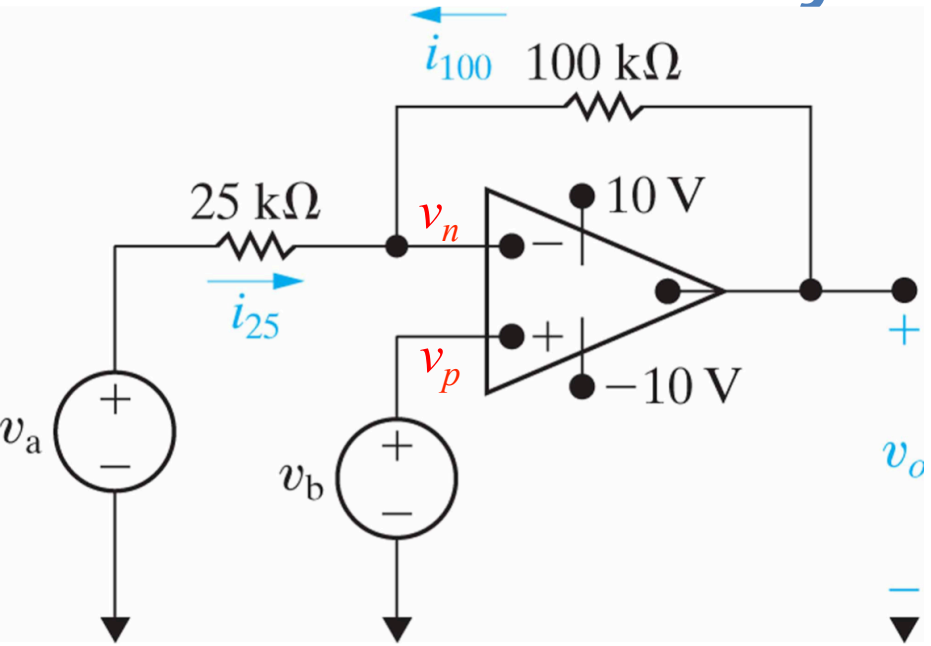
The op amp in the circuit is ideal.

a) Calculate v_o, if $v_a = 1$ V and $v_b = 0$ V.

b) If $v_a = 1.5$ V, specify the range of v_b that avoids amplifier saturation.



Solution for Example #1



a) $v_n = v_p = v_b = 0$

$$i_{25} = (v_a - v_n)/25 = \frac{1}{25} \text{ mA}$$

$$i_{100} = (v_o - v_n)/100 = v_o/100 \text{ mA}$$

$$i_{25} + i_{100} = i_n = 0$$

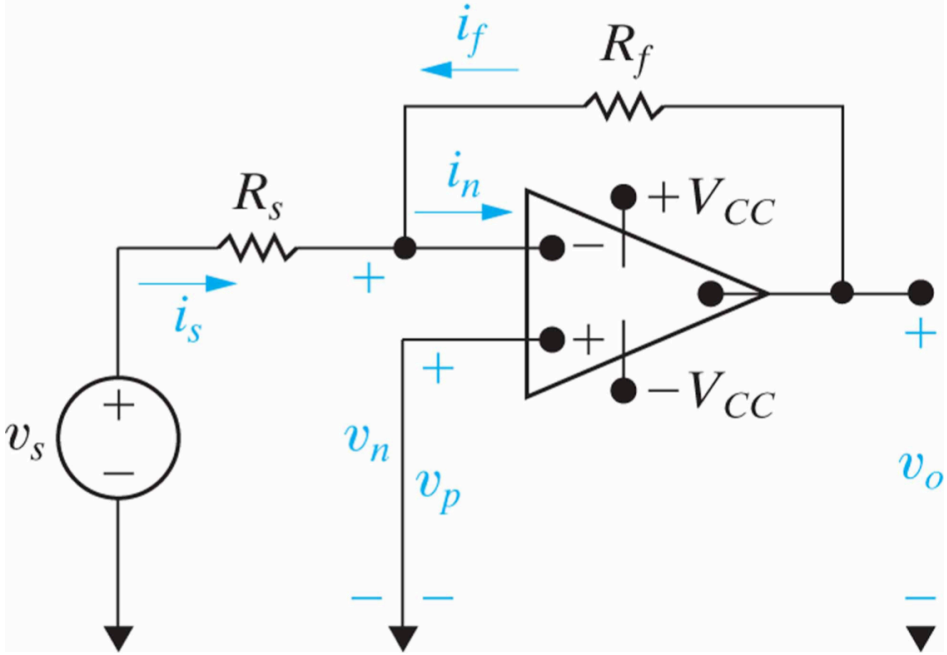
$$v_o = -4 \text{ V}$$

$$\frac{1.5 - v_b}{25} = -\frac{v_o - v_b}{100}$$

$$v_b = \frac{1}{5}(6 + v_o) \qquad -10 \text{ V} \le v_o \le 10 \text{ V}$$

$$-0.8 \text{ V} \le v_b \le 3.2 \text{ V}$$

The Inverting-Amplifier Circuit



$$v_n = v_p = 0$$

$$i_s + i_f = i_n = 0$$

$$i_s = \frac{v_s}{R_s} \qquad i_f = \frac{v_o}{R_f}$$

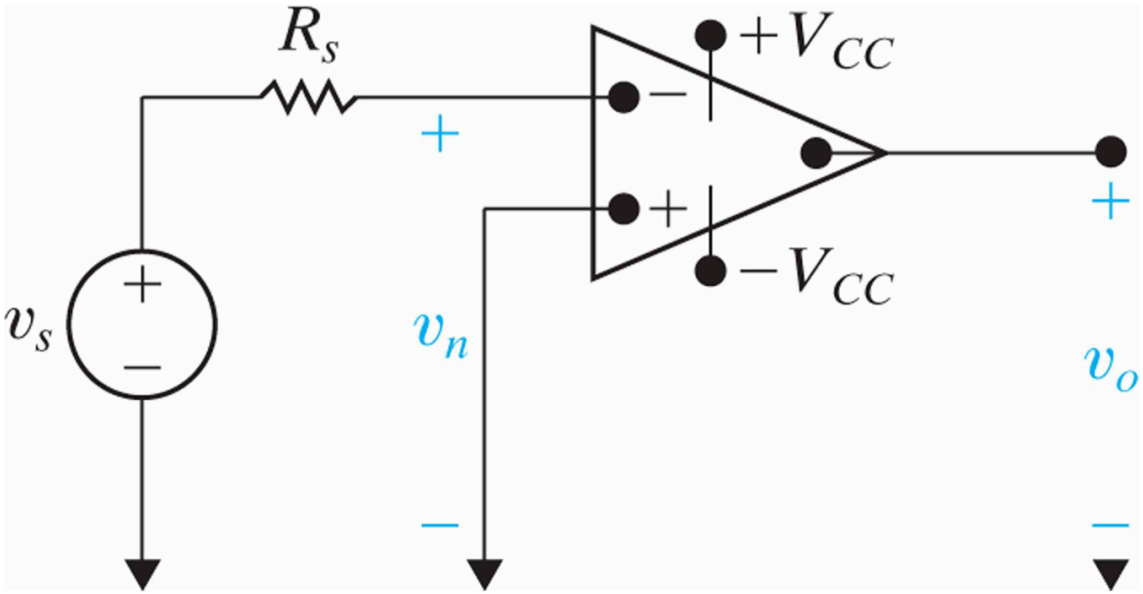
$$v_o = \frac{-R_f}{R_s} v_s$$

$$|v_o| \le V_{CC}$$

$$\left| \frac{R_f}{R_s} v_s \right| \le V_{CC}$$

$$\frac{R_f}{R_s} \le \left| \frac{V_{CC}}{v_s} \right|$$

Open-loop Gain



$$v_o = -Av_n = -Av_s$$

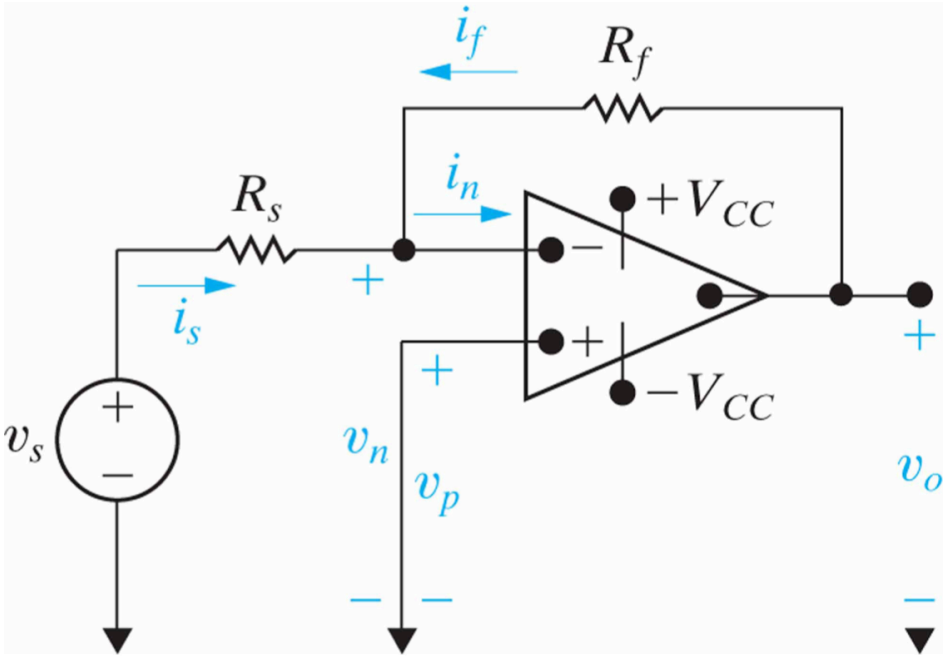
A: open-loop gain

$$|v_s| \le V_{CC}/A$$

Example #2

a) Design an inverting amplifier with a gain of 12. Use ±15 V power supplies and an ideal op amp.

b) What range of input voltages, v_s, allows the op amp in this design to remain in its linear operating region?



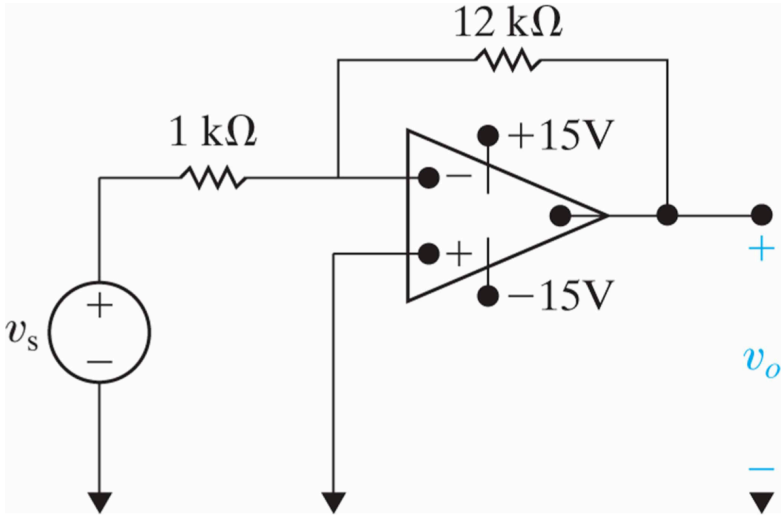
Solution for Example #2

a) We need to find two resistors whose **ratio is 12** from the realistic resistor values listed in Appendix H.

$$R_s = 1\ \text{k}\Omega \quad R_f = 12\ \text{k}\Omega$$

verify

$$v_o = -\frac{R_f}{R_s} v_s = -\frac{12{,}000}{1000} v_s = -12 v_s$$

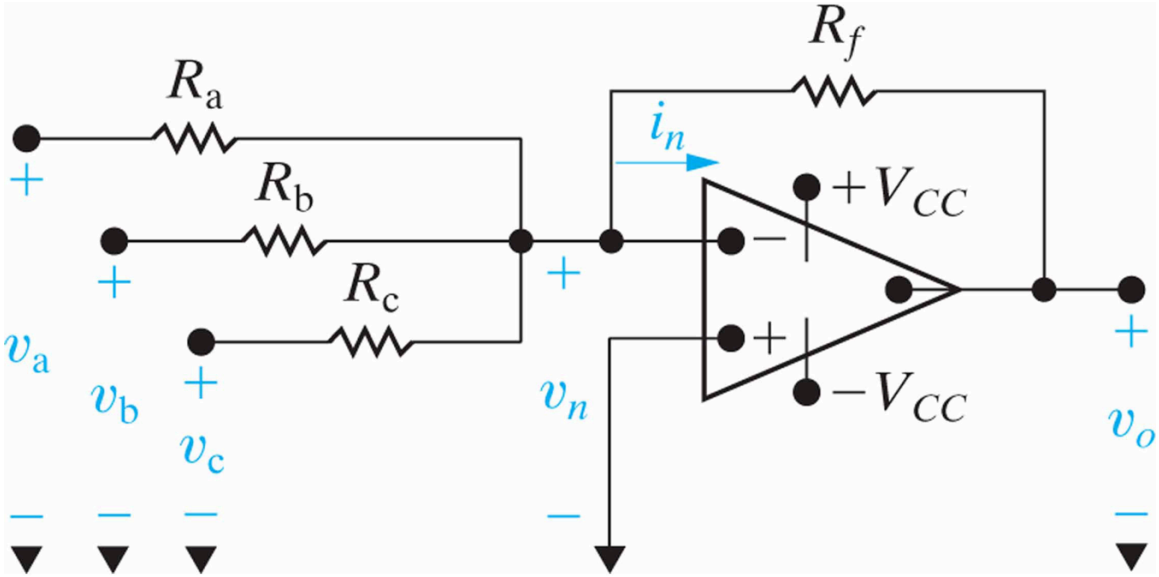


b) Solve two different versions of the inverting-amplifier equation for v_s: first using $v_o = +15$ V and then using $v_o = -15$ V:

$$15 = -12 v_s \quad \text{so} \quad v_s = -1.25\ \text{V}$$

$$-15 = -12 v_s \quad \text{so} \quad v_s = 1.25\ \text{V}$$

The Summing-Amplifier Circuit



$$R_a = R_b = R_c = R_s$$

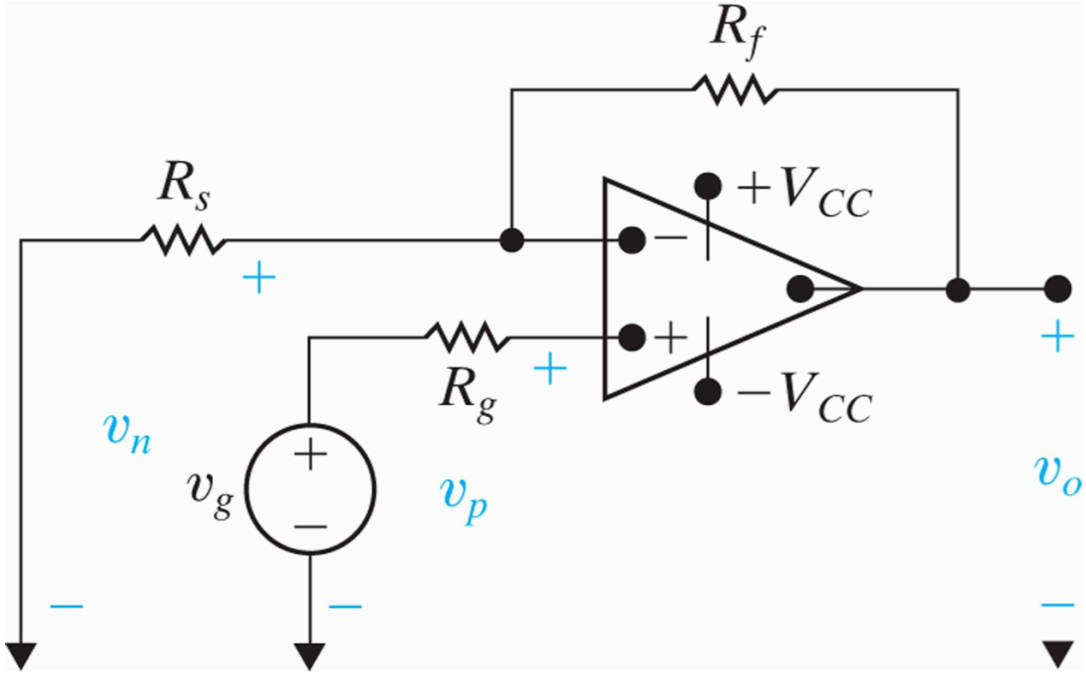
$$\frac{v_n - v_a}{R_a} + \frac{v_n - v_b}{R_b} + \frac{v_n - v_c}{R_c} + \frac{v_n - v_o}{R_f} + i_n = 0$$

$$i_n = 0 \qquad v_n = v_p = 0$$

$$v_o = -\left(\frac{R_f}{R_a}v_a + \frac{R_f}{R_b}v_b + \frac{R_f}{R_c}v_c\right)$$

$$v_o = -\frac{R_f}{R_s}(v_a + v_b + v_c)$$

The Noninverting-Amplifier Circuit



$$v_n = v_g = \frac{v_o R_s}{R_s + R_f}$$

$$v_o = \frac{R_s + R_f}{R_s} v_g$$

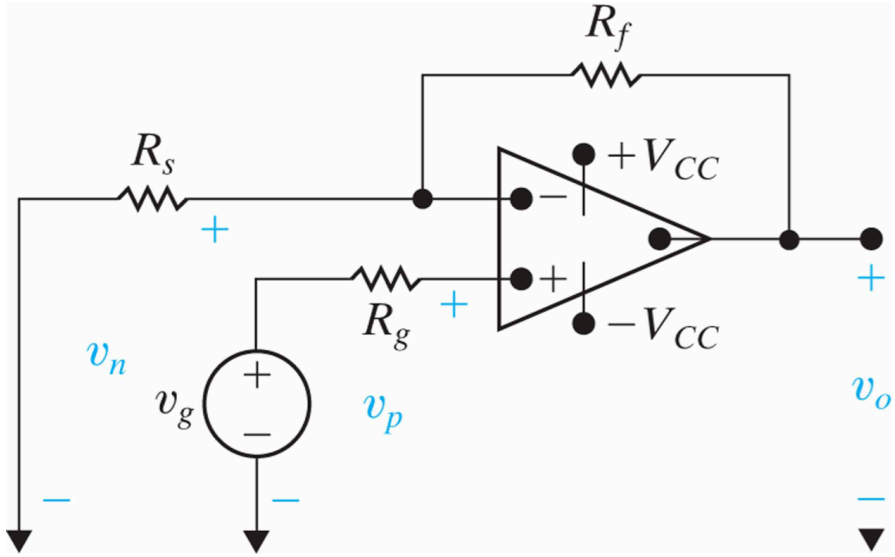
Operation in the linear region requires that

$$\frac{R_s + R_f}{R_s} < \left| \frac{V_{CC}}{v_g} \right|$$

Example #3

a) Design a noninverting amplifier with a gain of 6. Assume the op amp is ideal.

b) Suppose we wish to amplify a voltage v_g, such that $-1.5\ \text{V} \leq v_g \leq 1.5\ \text{V}$. What are the smallest power supply voltages that could be used with the resistors selected in part (a) and still have the op amp in this design remain in its linear operating region?



Solution for Example #3

a) Using the noninverting amplifier equation

$$v_o = \frac{R_s + R_f}{R_s} v_g = 6v_g \quad \text{so} \quad \frac{R_s + R_f}{R_s} = 6$$

Therefore, we have

$$R_s + R_f = 6R_s, \quad \text{so} \quad R_f = 5R_s$$

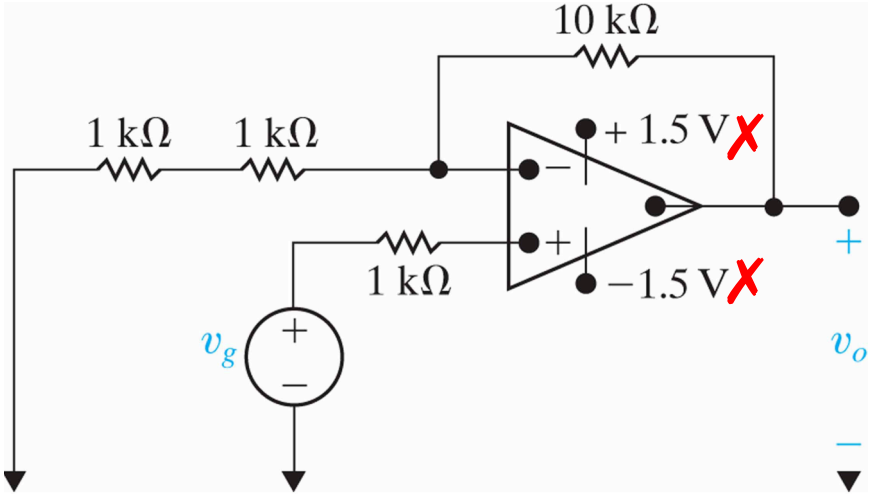
We want two resistors whose ratio is 5. Look at the realistic resistor values listed in Appendix H. Let's choose $R_f = 10\ \text{k}\Omega$, so $R_s = 2\ \text{k}\Omega$. But there is not a $2\ \text{k}\Omega$ resistor in Appendix H. We can create an equivalent $2\ \text{k}\Omega$ resistor by combining two $1\ \text{k}\Omega$ resistors in series. We can use a third $1\ \text{k}\Omega$ resistor as the value of the resistor R_g.

b) Solve two different versions of the noninverting amplifier equation for v_o—first using $v_g = +1.5$ V and then using $v_g =$ -1.5V:

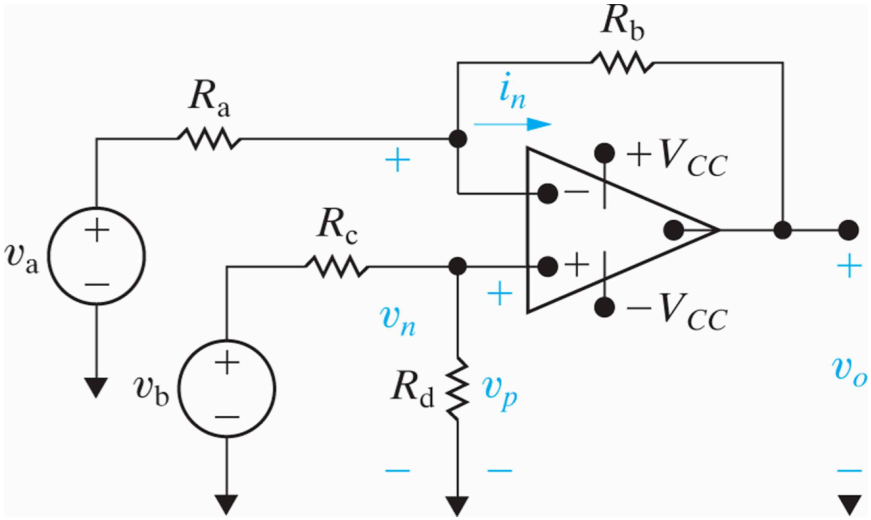
$$v_o = 6(1.5) = 9 \text{ V}$$

$$v_o = 6(-1.5) = -9 \text{ V}$$

Thus, if we use ± 9 V power supplies for the noninverting amplifier designed in part (a) and $-1.5 \text{ V} \leq v_g \leq +1.5$ V, the op amp will remain in its linear operating region.



The Difference-Amplifier Circuit



$$v_n = v_p = \frac{R_d}{R_c + R_d} v_b$$

$$i_n = i_p = 0$$

$$\frac{v_n - v_a}{R_a} + \frac{v_n - v_o}{R_b} + i_n = 0$$

$$\frac{R_a}{R_b} = \frac{R_c}{R_d}$$

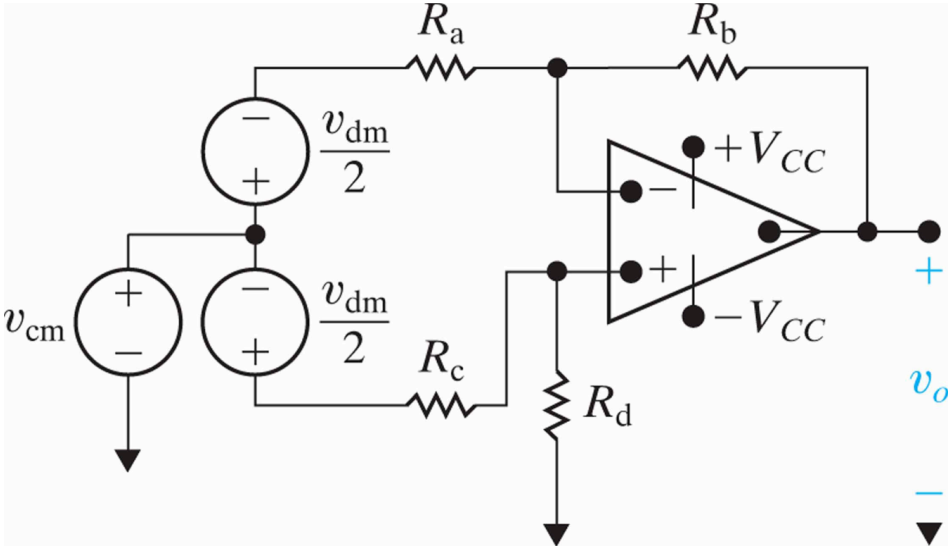
$$v_o = \frac{R_d(R_a + R_b)}{R_a(R_c + R_d)} v_b - \frac{R_b}{R_a} v_a$$

$$v_o = \frac{R_b}{R_a}(v_b - v_a)$$

The Difference Amplifier—Another Perspective

$$v_{dm} = v_b - v_a$$

$$v_{cm} = (v_a + v_b)/2$$



$$v_a = v_{cm} - \frac{1}{2}v_{dm}$$

$$v_b = v_{cm} + \frac{1}{2}v_{dm}$$

$$v_o = \left[\frac{R_a R_d - R_b R_c}{R_a(R_c + R_d)}\right]v_{cm}$$

$$+ \left[\frac{R_d(R_a + R_b) + R_b(R_c + R_d)}{2R_a(R_c + R_d)}\right]v_{dm},$$

$$= A_{cm}v_{cm} + A_{dm}v_{dm},$$

Insights

- In many applications it is the *differential mode signal that contains the information of interest*, whereas *the common mode signal is the noise* found in all electric signals.

- Thus, we hope

$$R_a R_d = R_b R_c$$



$$v_o = (0)v_{cm} + \left(\frac{R_b}{R_a}\right)v_{dm}$$

Common Mode Rejection Ratio

- An ideal difference amplifier has zero common mode gain and nonzero (and usually large) differential mode gain.
- Two factors have an influence on the ideal common mode gain—resistance mismatches or a nonideal op amp.

$$\frac{R_a}{R_b} = (1 - \epsilon)\frac{R_c}{R_d}$$

$$R_a = (1 - \epsilon)R_c \quad \text{and} \quad R_b = R_d$$

or

$$R_d = (1 - \epsilon)R_b \quad \text{and} \quad R_a = R_c$$

$$A_{cm} = \frac{R_a(1-\epsilon)R_b - R_aR_b}{R_a[R_a + (1-\epsilon)R_b]}$$

$$= \frac{-\epsilon R_b}{R_a + (1-\epsilon)R_b}$$

$$\approx \frac{-\epsilon R_b}{R_a + R_b}.$$

$$A_{dm} = \frac{(1-\epsilon)R_b(R_a + R_b) + R_b[R_a + (1-\epsilon)R_b]}{2R_a[R_a + (1-\epsilon)R_b]}$$

$$= \frac{R_b}{R_a}\left[1 - \frac{(\epsilon/2)R_a}{R_a + (1-\epsilon)R_b}\right]$$

$$\approx \frac{R_b}{R_a}\left[1 - \frac{(\epsilon/2)R_a}{R_a + R_b}\right].$$

The **common mode rejection ratio (CMRR)** can be used to measure how nearly ideal a difference amplifier is. It is defined as the ratio of the differential mode gain to the common mode gain

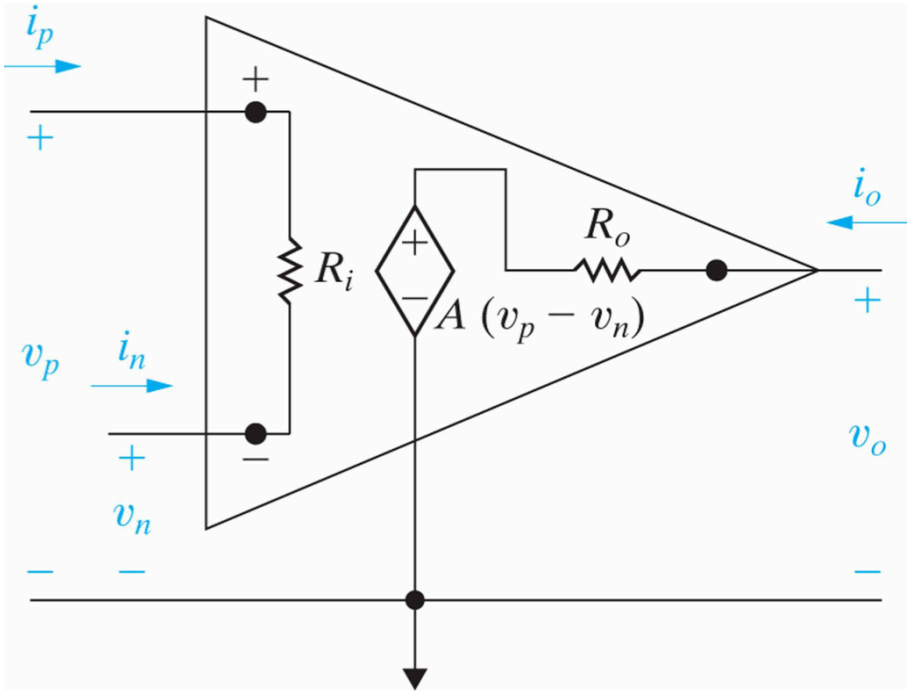
$$\text{CMRR} = \left|\frac{A_{dm}}{A_{cm}}\right| \approx \left|\frac{1 + R_b/R_a}{-\epsilon}\right|$$

A More Realistic Model for the Operational Amplifier

A realistic model includes three modifications to the ideal op amp: (1) a finite input resistance, R_i; (2) a finite open-loop gain, A; and (3) a nonzero output resistance, R_o.

$$v_n = v_p$$ ✗

$$i_n = i_p = 0$$ ✗



For the juA741 op amp, the typical values of R_i, A, and R_o are 2 MΩ, 10^5, and 75 Ω, respectively.

Analysis of the More Realistic Op Amp Model

- Take the Inverting-Amplifier Circuit for example:

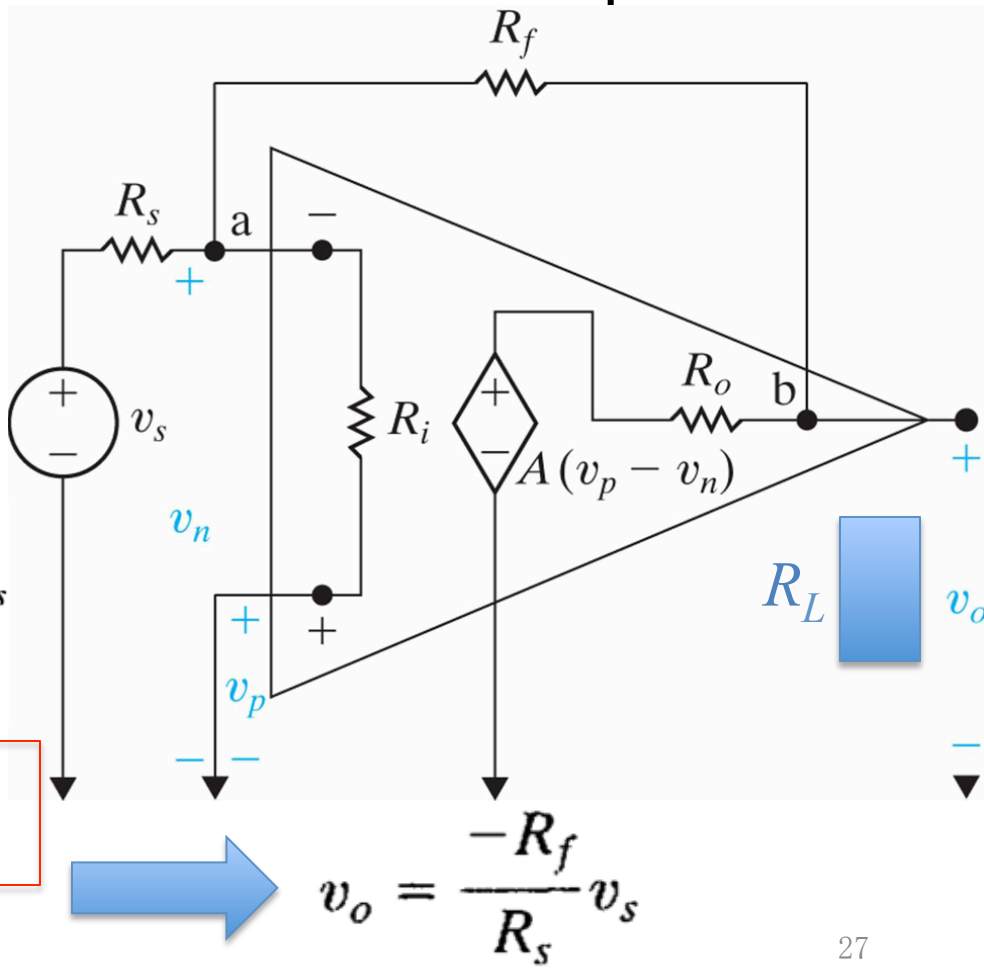
$$\text{node a:}\quad \frac{v_n - v_s}{R_s} + \frac{v_n}{R_i} + \frac{v_n - v_o}{R_f} = 0$$

$$\text{node b:}\quad \frac{v_o - v_n}{R_f} + \frac{v_o - A(-v_n)}{R_o} = 0$$

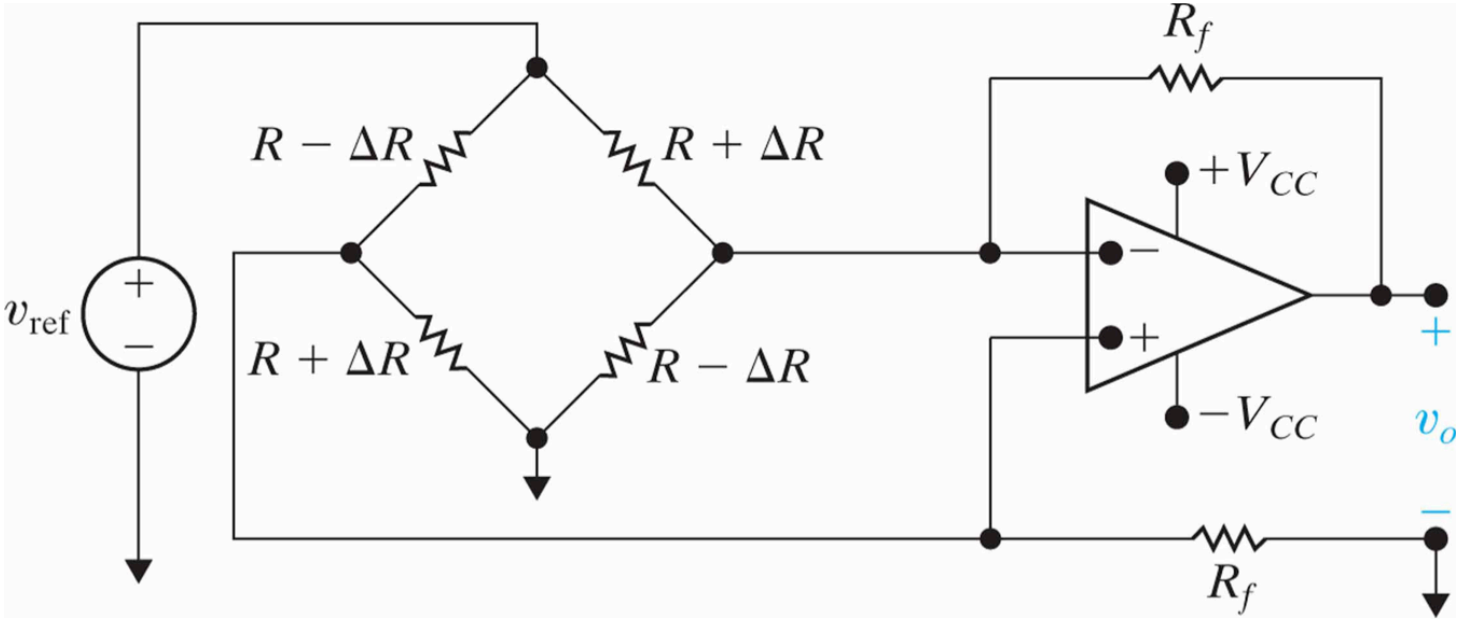
$$v_o = \frac{-A + (R_o/R_f)}{\frac{R_s}{R_f}\left(1 + A + \frac{R_o}{R_i}\right) + \left(\frac{R_s}{R_i} + 1\right) + \frac{R_o}{R_f}} v_s$$

$$R_i \to \infty,\ A \to \infty,\ \text{and}\ R_o \to 0$$

$$v_o = \frac{-R_f}{R_s} v_s$$



Strain Gages



The pair of strain gages that are lengthened once the bar is bent have the values $R + \Delta R$ in the bridge feeding the difference amplifier, whereas the pair of strain gages that are shortened have the values $R - \Delta R$.

$$\frac{v_{\text{ref}} - v_n}{R + \Delta R} = \frac{v_n}{R - \Delta R} + \frac{v_n - v_o}{R_f}$$

$$\frac{v_{\text{ref}} - v_p}{R - \Delta R} = \frac{v_p}{R + \Delta R} + \frac{v_p}{R_f}$$

$$v_p = \frac{v_{\text{ref}}}{(R - \Delta R)\left(\frac{1}{R + \Delta R} + \frac{1}{R - \Delta R} + \frac{1}{R_f}\right)}$$

$$v_o = \frac{R_f(2\Delta R)}{R^2 - (\Delta R)^2} v_{\text{ref}}$$

$$(\Delta R)^2 \ll R^2, \text{ so } R^2 - (\Delta R)^2 \approx R^2$$

$$\delta = \Delta R / R$$

$$v_o \approx \frac{R_f}{R} 2\delta v_{\text{ref}}$$

Summary

- Ideal op amp, linear region, saturate
- Voltage constraint, current constraint
- Inverting amplifier, summing amplifier, noninverting amplifier, difference amplifier
- Common mode and difference mode
- Difference-amplifier, common mode rejection ratio (CMRR)
- More realistic Op Amp model